Cambridge Assessment International Education

Cambridge International AS & A Level

MATHEMATICS **9709/52**

Paper 5 Probability & Statistics 1 **May/June 2021**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mathematics Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

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Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

DM or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

FT Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

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Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

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Question	Answer	Marks	Guidance
1(a)	6	B1	WWW
		1	
1(b)	$\left(\frac{5}{6}\right)^3\frac{1}{6}+\left(\frac{5}{6}\right)^4\frac{1}{6}+\left(\frac{5}{6}\right)^5\frac{1}{6}+\left(\frac{5}{6}\right)^6\frac{1}{6}$	M1	$p^3(1-p)+p^4(1-p)+p^5(1-p)+p^6(1-p)$, $0<p<1$
	0.300 (0.2996…)	A1	At least 3s.f. Award at most accurate value.
	Alternative method for Question 1(b)		
	$\left(\frac{5}{6}\right)^3-\left(\frac{5}{6}\right)^7$	M1	p^3-p^7, $0<p<1$
	0.300 (0.2996…)	A1	At least 3s.f. Award at most accurate value.
		2	
1(c)	$1-\left(\frac{5}{6}\right)^9$	M1	$1-p^n$, $0<p<1$, $n=9, 10$
	0.806	A1	
	Alternative method for Question 1(c)		
	$\frac{1}{6}+\frac{1}{6}\left(\frac{5}{6}\right)+\frac{1}{6}\left(\frac{5}{6}\right)^2+\cdots+\frac{1}{6}\left(\frac{5}{6}\right)^8$	M1	$p+p(1-p)+p(1-p)^2+p(1-p)^3+p(1-p)^4+p(1-p)^5+p(1-p)^6+p(1-p)^7+p(1-p)^8(+p(1-p)^9)$, $0<p<1$ As per answer for minimum terms shown
	0.806	A1	
		2	

Question	Answer	Marks	Guidance
2	$\left[P(X>1.1)=\frac{72}{2000}(=0.036)\right]$ $z=\pm 1.798$	B1	$1.79 < z \leqslant 1.80$, $-1.80 \leqslant z < -1.79$ seen
	$\frac{1.1-1.04}{\sigma}=1.798$ $\left[\frac{0.06}{\sigma}=1.798\right]$	B1	1.1 and 1.04 substituted in ±standardisation formula, allow continuity correction, not σ^2 or $\sqrt{\sigma}$
		M1	Equate *their* ±standardisation formula to a z-value and to solve for the appropriate area leading to final answer (expect $\sigma < 0.5$). $\left(\text{Accept} \pm \frac{0.06}{\sigma} = z - \text{value}\right)$
	$\sigma = 0.0334$	A1	$0.03335 \leq \sigma \leq 0.0334$. At least 3 3s.f.
		4	

Question	Answer	Marks	Guidance
3(a)	P(not late) = 0.4 × 0.45 + 0.35 × 0.3 + 0.25 × (1 – x) or P(late) = 0.4 × 0.55 + 0.35 × 0.7 + 0.25x	M1	$0.4\times p+0.35\times q+0.25\times r$, p = 0.45, 0.55, q = 0.3, 0.7 and r = (1 – x), x
	0.18 + 0.105 + 0.25 (1 – x) = 0.48 or 0.22 + 0.245 + 0.25x = 0.52	A1	Linear equation formed using sum of 3 probabilities and 0.48 or 0.52 as appropriate. Accept unsimplified.
	x = 0.22	A1	Final answer
		3	
3(b)	$\left[P\left(train\middle\vert late\right)=\frac{P\left(train\cap late\right)}{P\left(late\right)}\right]$ $=\frac{0.35\times0.7}{1-0.48}$ or $\frac{0.35\times0.7}{0.4\times0.55+0.35\times0.7+0.25\times their\,0.22}$	B1	0.35×0.7 or 0.245 seen as numerator of fraction
		M1	P(late) seen as a denominator with *their* probability as numerator (Accept $\frac{their\ p}{0.52}$ or $\frac{their\ p}{0.22+0.245+0.25\times their\,0.22}$)
	= 0.471 or $\frac{49}{104}$	A1	
		3	

Question	Answer	Marks	Guidance
4(a)	X: –1, 0, 1, 2, 3 $P(X)$: $\frac{1}{9}$, $\frac{2}{9}$, $\frac{1}{9}$, $\frac{3}{9}$, $\frac{2}{9}$	B1	Table with correct X values and at least one probability Condone any additional X values if probability stated as 0.
		B1	2 correct probabilities linked with correct outcomes, may not be in table.
		B1	3 further correct probabilities linked with correct outcomes, may not be in table. **SC** if less than 2 correct probabilities seen, award **SCB1** for **s**um of *their* 4 or 5 probabilities in table = 1
		3	
4(b)	$\left[E(X)=\frac{-1\times1+(0\times2)+1\times1+2\times3+3\times2}{9}=\right]$ $\frac{-1+1+6+6}{9}$	M1	May be implied by use in variance, accept unsimplified expression. FT *their* table if *their* 3 or more probabilities sum to 1 or 0.999
	[Var (X) =] $\left[\frac{-1^2\times1+(0^2\times2)+1^2\times1+2^2\times3+3^2\times2}{9}-(their\ E(X))^2\right]$ $\frac{1+0+1+12+18}{9}-(their\ E(X))^2$	M1	Appropriate variance formula using *their* $(E(X))^2$ value. FT *their* table even if *their* 3 or more probabilities not summing to 1.
	$E(X)=\frac{4}{3}$ or 1.33 and $Var(X)=\frac{16}{9}$ or 1.78	A1	Answers for E(X) and Var(X) must be identified
		3	**N.B.** If method FT for M marks from *their* incorrect **(b)**, expressions for E(X) and Var(X) must be seen unsimplified with all probabilities <1

Question	Answer	Marks	Guidance
5(a)	$[(0.7)^3 =]0.343$	B1	Evaluated WWW
	Alternative method for Question 5(a)		
	$[(0.15)^3 + {}^3C_1(0.15)^2(0.55) + {}^3C_2(0.15)(0.55)^2 + (0.55)^3 =] 0.343$	B1	Evaluated WWW
		1	
5(b)	$1 - (0.85^9 + {}^9C_1\, 0.15^1\, 0.85^8 + {}^9C_2\, 0.15^2\, 0.85^7)$ $[1 - (0.231617 + 0.367862 + 0.259667)]$	M1	One term: ${}^9C_x\, p^x\, (1 - p)^{9-x}$ for $0 < x < 9$, any $0 < p < 1$
		A1	Correct expression, accept unsimplified.
	0.141	A1	$0.1408 \leqslant$ ans $\leqslant 0.141$, award at most accurate value.
	Alternative method for Question 5(b)		
	${}^9C_3\, 0.15^3\, 0.85^6 + {}^9C_4\, 0.15^4\, 0.85^5 + {}^9C_5\, 0.15^5\, 0.85^4 + {}^9C_6\, 0.15^6\, 0.85^3 +$ ${}^9C_7\, 0.15^7\, 0.85^2 + {}^9C_8\, 0.15^8\, 0.85 + 0.15^9$	M1	One term: ${}^9C_x\, p^x\, (1 - p)^{9-x}$ for $0 < x < 9$, any $0 < p < 1$
		A1	Correct expression, accept unsimplified.
	0.141	A1	$0.1408 \leqslant$ ans $\leqslant 0.141$, award at most accurate value.
		3	

Question	Answer	Marks	Guidance
5(c)	Mean $=[60\times0.15=]9$ Variance $=[60\times0.15\times0.85=]7.65$	B1	Correct mean and variance, allow unsimplified. ($2.765 \le \sigma \le 2.77$ imply correct variance)
	$\left[(X\ge12)=\right]P\left(Z>\dfrac{11.5-9}{\sqrt{7.65}}\right)$	M1	Substituting *their* mean and variance into ±standardisation formula (any number for 11.5), not σ^2 or $\sqrt{\sigma}$
		M1	Using continuity correction 11.5 or 12.5 in *their* standardisation formula.
	$1-\Phi(0.9039)=1-0.8169$	M1	Appropriate area Φ, from final process, must be probability.
	0.183	A1	Final AWRT
		5	

Question	Answer	Marks	Guidance
6(a)	$\dfrac{8!}{2!3!}$	M1	$\dfrac{8!}{k!\times m!}$ $k = 1$ or 2, $m = 1$ or 3, not $k = m = 1$ no additional terms
	3360	A1	
		2	

Question	Answer	Marks	Guidance
6(b)	**Method 1** Arrangements Rs at ends – Arrangements Rs at ends and Os together		
	[Os not together $=$] $\frac{6!}{3!} - 4!$	M1	$\frac{6!}{k!} - m$, $1 \leqslant k \leqslant 3$, m an integer, condone $2\times\left(\frac{6!}{k!}\right) - m$.
		M1	$w - 4!$ or $w - 24$, w an integer Condone $w - 2 \times 4!$
	96	A1	
	Method 2 identified scenarios R _ _ _ R, Arrangement No Os together + 2Os and a single O		
	${}^4C_3 \times 3! + {}^4C_2 \times 2 \times 3!$	M1	${}^4C_3 \times 3! + r$ or $4\times 3! + r$ or ${}^4P_3 \times 3! + r$, r an integer. Condone $2 \times {}^4C_3 \times 3! + r$. $2 \times 4\times 3! + r$ or $2 \times {}^4P_3 \times 3! + r$.
		M1	$q + {}^4C_2 \times 3! \times k$ or $q + {}^4P_2 \times 3! \times k$, $k = 1,2$, q an integer
	[24 + 72 =] 96	A1	
		3	
6(c)	**Method 1** Identified scenarios		
	OORR ${}^3C_2 \times {}^2C_2 \times \left[{}^3C_0\right] = 3\times1 = 3$ ORR_ ${}^3C_1 \times {}^2C_2 \times {}^3C_1 = 3\times1\times3 = 9$ OOR_ ${}^3C_2 \times {}^2C_1 \times {}^3C_1 = 3\times2\times3 = 18$ OR_ _ ${}^3C_1 \times {}^2C_1 \times {}^3C_2 = 3\times2\times3 = 18$ OOOR ${}^3C_3 \times {}^2C_1 \times \left[{}^3C_0\right] = 1\times2 = 2$	B1	Outcomes for 2 identifiable scenarios correct, accept unsimplified.
		M1	Add 4 or 5 identified correct scenarios only values, no additional incorrect scenarios, no repeated scenarios, accept unsimplified, condone use of permutations.
	Total 50	A1	All correct and added
	Probability $= \frac{50}{{}^8C_4}$	M1	$\frac{\textit{their}'50'}{{}^8C_4}$, accept numerator unevaluated

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Question	Answer	Marks	Guidance
6(c) cont'd	$\frac{50}{70}$ or 0.714	A1	
	Method 2 Identified outcomes		
	ORTM $^3C_1 \times {}^2C_1 = 6$ ORTW $^3C_1 \times {}^2C_1 = 6$	B1	Outcomes for 5 identifiable scenarios correct, accept unsimplified.
	ORMW $^3C_1 \times {}^2C_1 = 6$ ORRM $^3C_1 \times {}^2C_2 = 3$ ORRW $^3C_1 \times {}^2C_2 = 3$ ORRT $^3C_1 \times {}^2C_2 = 3$ OROR $^3C_2 \times {}^2C_2 = 3$ OROT $^3C_2 \times {}^2C_1 = 6$ OROM $^3C_2 \times {}^2C_1 = 6$ OROW $^3C_2 \times {}^2C_1 = 6$ OROO $^3C_3 \times {}^2C_1 = 2$	M1	Add 9, 10 or 11 identified correct scenarios only values, no additional incorrect scenarios, no repeated scenarios, accept unsimplified, condone use of permutations.
	Total 50	A1	All correct and added
	Probability = $\frac{50}{{}^8C_4}$	M1	$\frac{\textit{their}'50'}{{}^8C_4}$, accept numerator unevaluated.
	$\frac{50}{70}$ or 0.714	A1	
		5	

Question	Answer	Marks	Guidance
7(a)	Includes all data	B1	Reference to *either* including all/raw data or further statistical processes are possible that cannot be found using data from box-and-whisker, eg frequency, mean, mode or standard deviation **not** only median, IQR, range or spread which can be found from both.
		1	
7(b)	<table><tr><td>Amazons</td><td></td><td>Giants</td></tr><tr><td>8</td><td>17</td><td>5</td></tr><tr><td>4 2 1</td><td>18</td><td>2 4 7 9</td></tr><tr><td>8 6 0</td><td>19</td><td>2 3 5 5 5</td></tr><tr><td>5 2 1</td><td>20</td><td>4</td></tr><tr><td>5</td><td>21</td><td></td></tr></table> Key: 1|18|2 means 181 cm for Amazons and 182 cm for Giants	B1	Correct stem can be upside down, ignore extra values
		B1	Correct Amazons labelled on left, leaves in order from right to left and lined up vertically (less than halfway to next column), no commas or other punctuation.
		B1	Correct Giants labelled on same diagram, leaves in order and lined up vertically (less than halfway to next column), no commas or other punctuation.
		B1	Correct single key for their diagram, need both teams identified and 'cm' stated at least once here or in leaf headings or title. **SC** for if 2 separate diagrams drawn, award **SCB1** if both keys meet these criteria (Max B1, B0, B0, B1)
		4	
7(c)	[UQ = 202 (cm), LQ = 182 (cm)] [IQR =] 202 – 182 = 20 (cm)	M1	$201 \leqslant UQ \leqslant 205 - 181 \leqslant LQ \leqslant 184$
		A1	WWW
		2	

Question	Answer	Marks	Guidance
7(d)	[$\Sigma_{11} = 2132$ $\Sigma_{15} = 191.2 \times 15 = 2868$]	B1	Both Σ_{11} and Σ_{15} found. Accept unevaluated.
	their 2868 = *their* 2132 + (180 + 185 + 190) + h	M1	Forming an equation for the height using *their* Σ_{11} and Σ_{15}.
	181 (cm)	A1	
	Alternative method for Question 7(d)		
	[$\Sigma_{15} = 191.2 \times 15 = 2868$ $\Sigma_{15} = 2687 + h$]	B1	Σ_{15} found using the mean and raw data methods. Accept unevaluated.
	their 2868 = *their* 2687 + h	M1	Forming an equation for the height using *their* Σ_{15} expressions.
	181 (cm)	A1	
	Alternative method for Question 7(d)		
	[$\Sigma_{15} = 2687 + h$ $\frac{\Sigma_{15}}{15} = 191.2$]	B1	Σ_{15} found using raw data method and statement on calculating new mean. Accept unevaluated.
	$\frac{\textit{their}\ 2687 + h}{15} = 191.2$	M1	Forming an equation for the height using *their* Σ_{15} expressions
	181 (cm)	A1	
		3	**N.B.** All methods can be presented as a logical numerical argument which can be condoned if clear.